AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 17 · STUDENT LESSON

Model Memory vs Retrieval

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

Name the evidence need. Choose a route. Preserve the gap.

Distinguish learned patterns from inspectable retrieved records and calculated results.

Keep one habit: Evidence contract → library notice → stale index → twelve routing cards → independent corpus task.

MAIN LESSON · SECTION 1

Where does this answer's evidence come from?

> Learned patterns, retrieved records and calculated values have different evidence trails.

Your learning goals

I can explain the **mechanism** and terminology of **retrieval** and **index**ing.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Recall source identity and calculation limits

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

- Recall Verify Numbers and Dates: a correct calculation does not verify its **input** source. Name which claim requires a current record and which can be computed from supplied facts.

Your end goal

By the end, I can explain model memory vs **retrieval** using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Verify Numbers and Dates. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Hallucination Failure Types. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Visual Composition. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 17: Model Memory vs **Retrieval**

Learned **parameters** are not a searchable source library.



CREATOR MISSION

Decide when to answer, retrieve, calculate, or **abstain**.

Creator name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

This week’s end goal

MAIN LESSON · SECTION 2

Separate learned patterns from inspectable records

> Fluent wording does not identify its source.

What memory and retrieval mean

Model memory is an informal name for patterns encoded in learned parameters. It is not a document catalogue that guarantees the exact origin, date or wording of an answer. Retrieval selects stored records for a query; a later generator may use those records to compose an answer. Retrieval makes a source inspectable, but selection, freshness, permissions and correct use still need checking.

This lesson uses supplied paper records, not a live model or search engine.

Understand the retrieval operation

Retrieval returns stored records selected for a query. An **index** is a searchable representation of a versioned **corpus**; generated wording is a later operation.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

▼ **Earlier wording, alternatives and picture transcripts**

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use model memory vs **retrieval** to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

MAIN LESSON · SECTION 3

Trace corpus → eligible index → passage → answer

> Document identity and version must survive the pipeline.

Set IDs, versions and permissions

Assign stable document IDs, versions and permitted access before **index**ing. Retrieve only authorised records. Preserve source references and define behaviour when a document is missing, outdated or access-restricted.

Apply the full source-selection contract

Classroom contract: the corpus is the collection of available records; the index is its searchable representation, which may lag behind it. Match document ID, version, access eligibility and requested time before using a retrieved passage. A high similarity score is a relevance clue, not permission or proof of freshness. If a current approved source is unavailable, state that the current fact cannot be verified.

Do not substitute an old generated answer.

Preserve missing-input limits

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Explain the deciding check

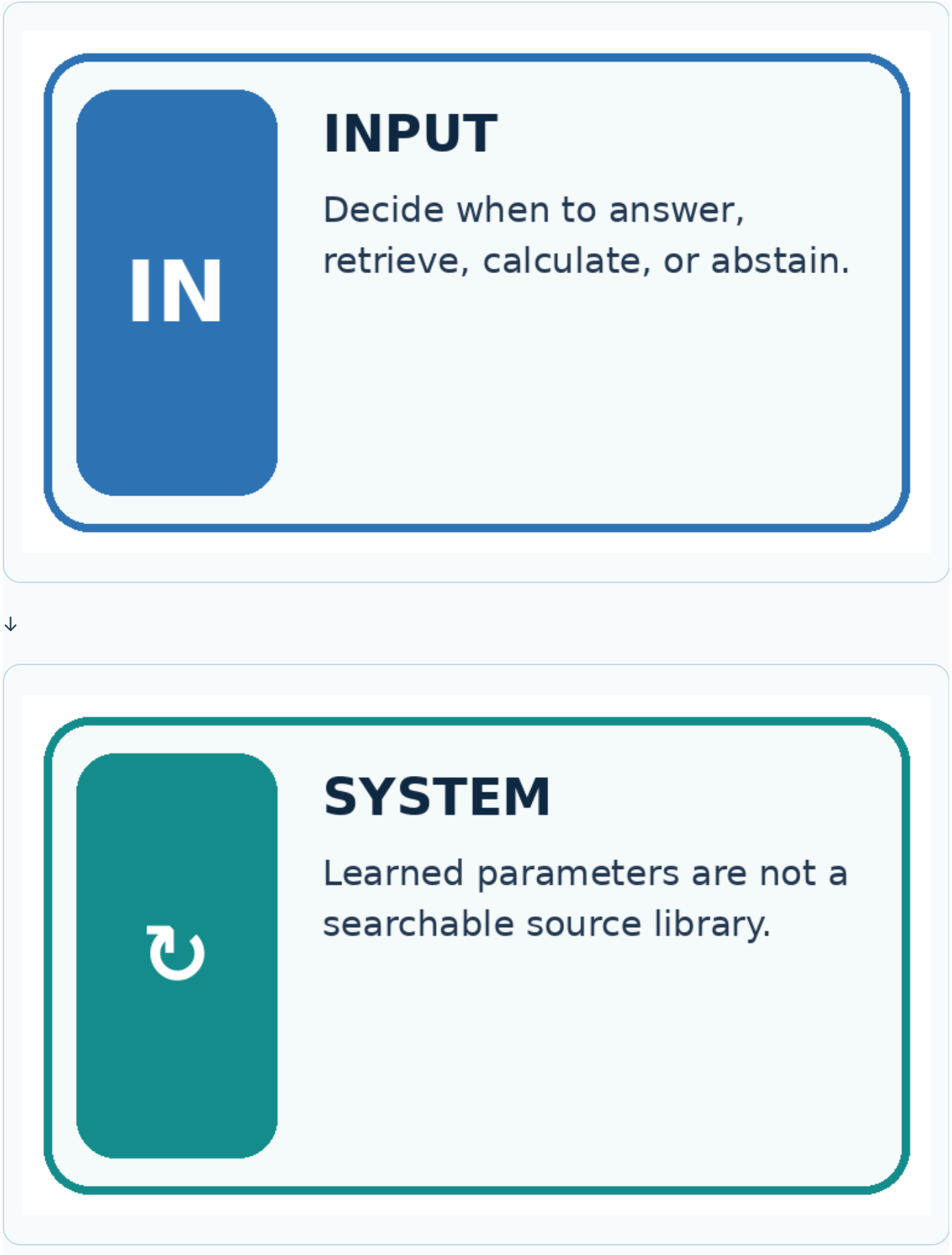
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternatives and picture transcripts

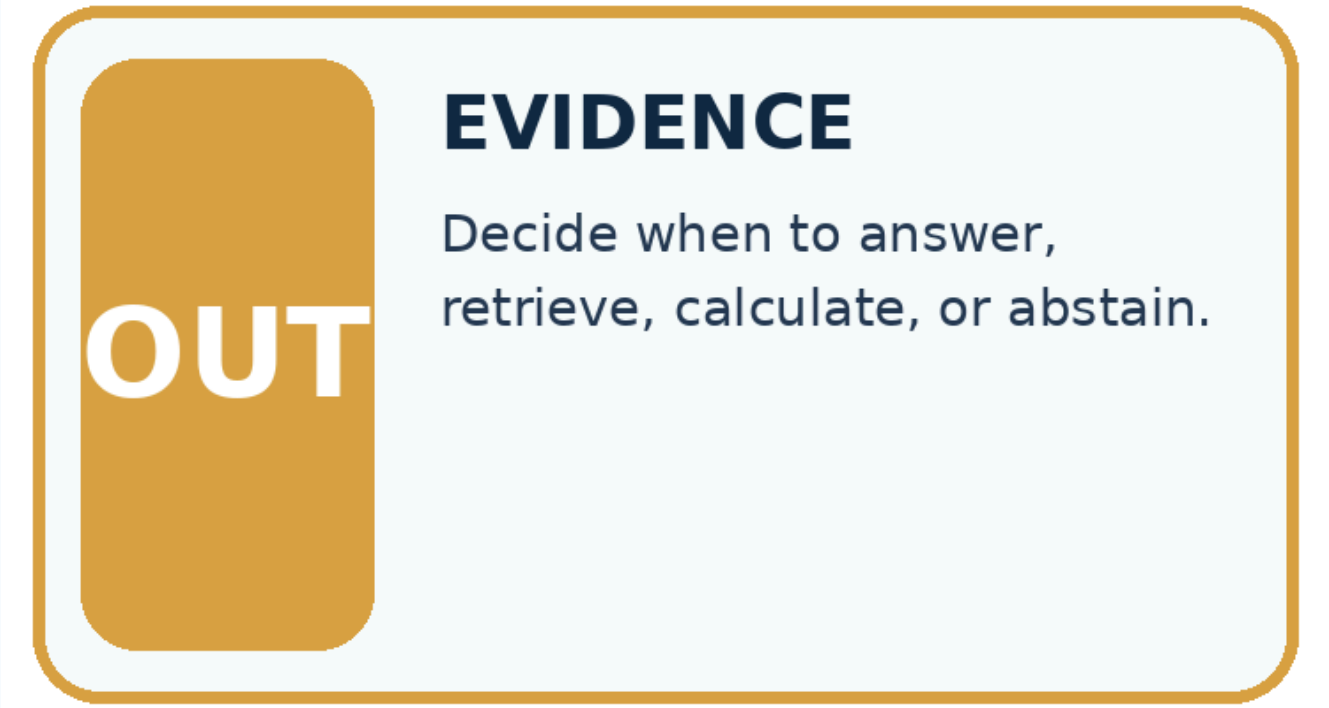
VISUAL **MECHANISM**

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.


IN
INPUT
Decide when to answer, retrieve, calculate, or abstain.
↓
SYSTEM
Learned parameters are not a searchable source library.

↓



Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Retrieval and indexing — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Index v1 contains notice N1 for room A. Corpus v2 supersedes it with N2 for room B, but the query still returns N1. What can be reported?	Assign stable document IDs, versions and permitted access before indexing. Retrieve only authorised records. Preserve source references and define behaviour when a document is missing, outdated or access-restricted.	The stale index returns old evidence. It cannot establish current room approval; refresh against authorised corpus v2 or report unresolved freshness.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

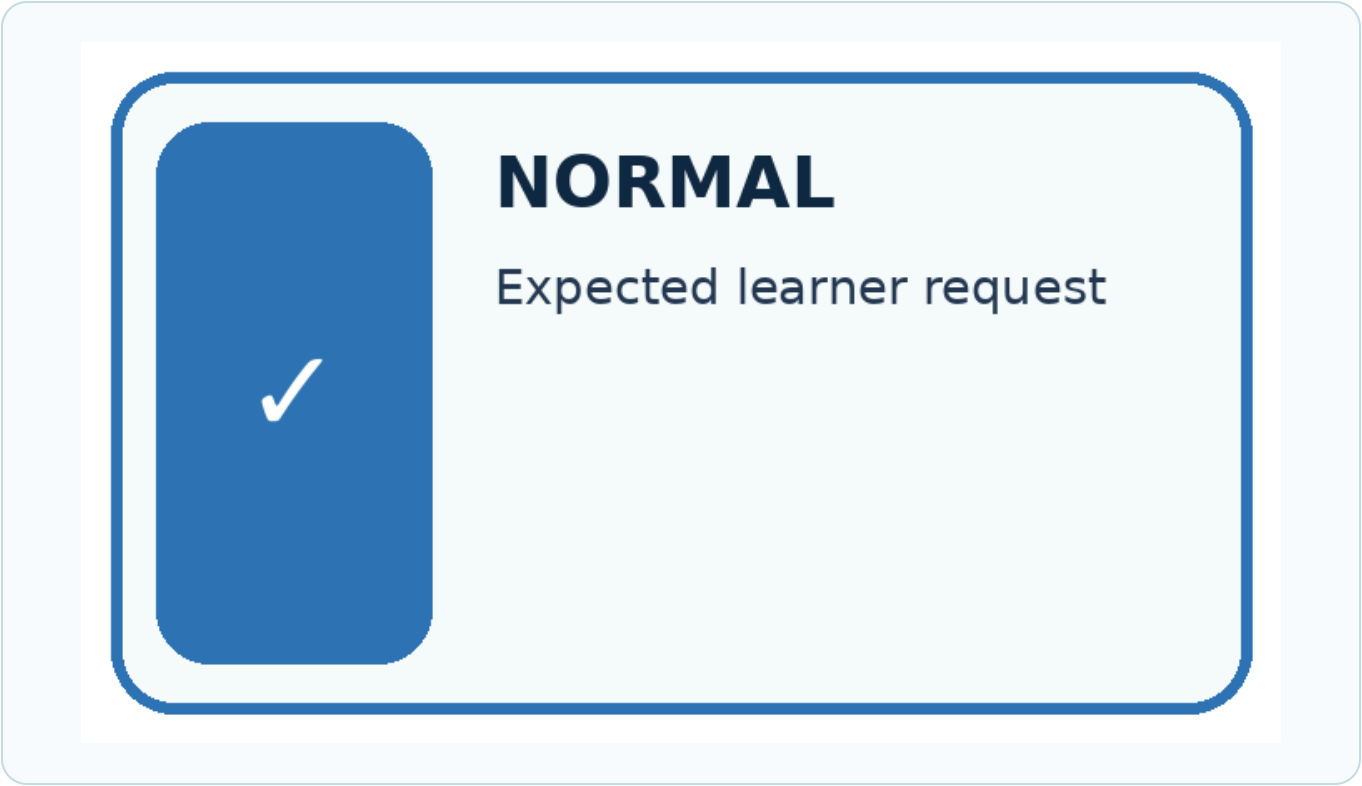
Connect the picture and the explanation

MAIN LESSON · SECTION 4

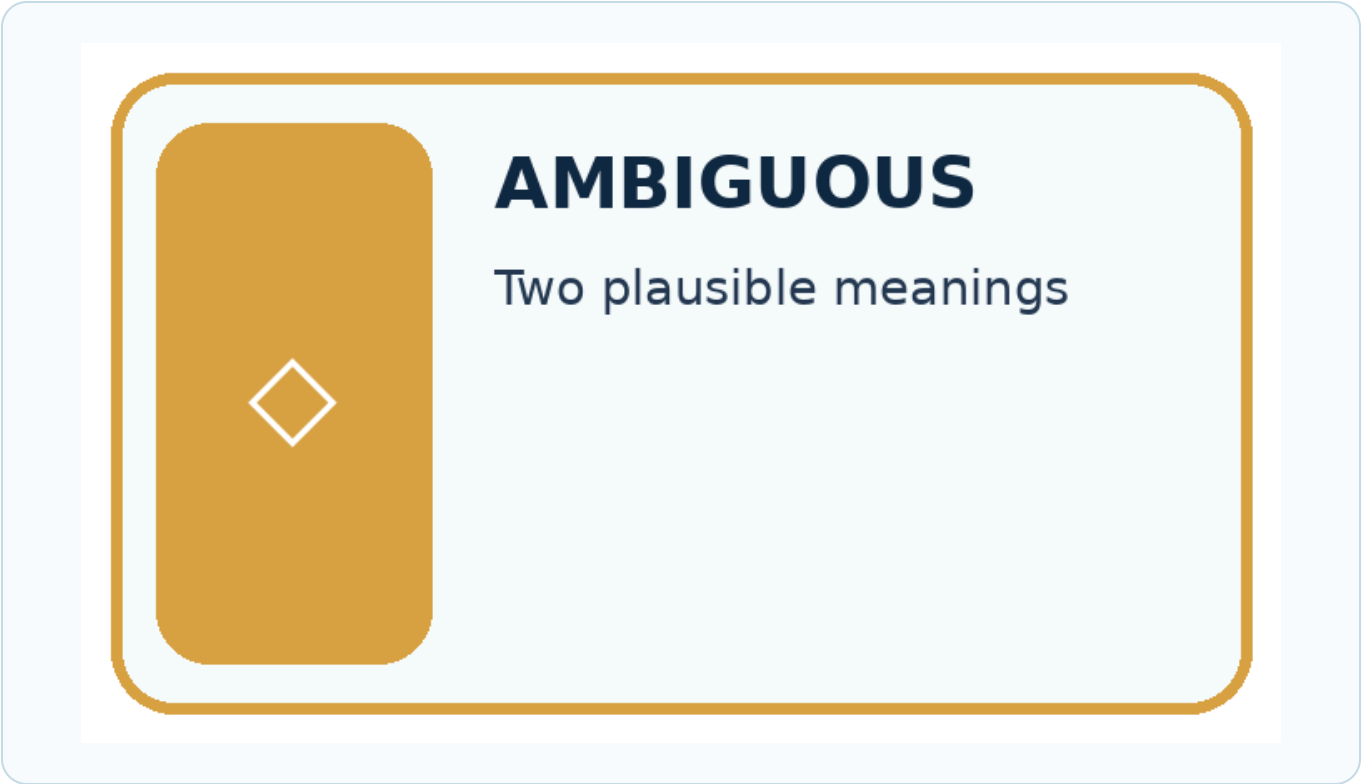
Compare available evidence with plausible wording

> A retrieval failure does not make an unsupported old answer reliable.

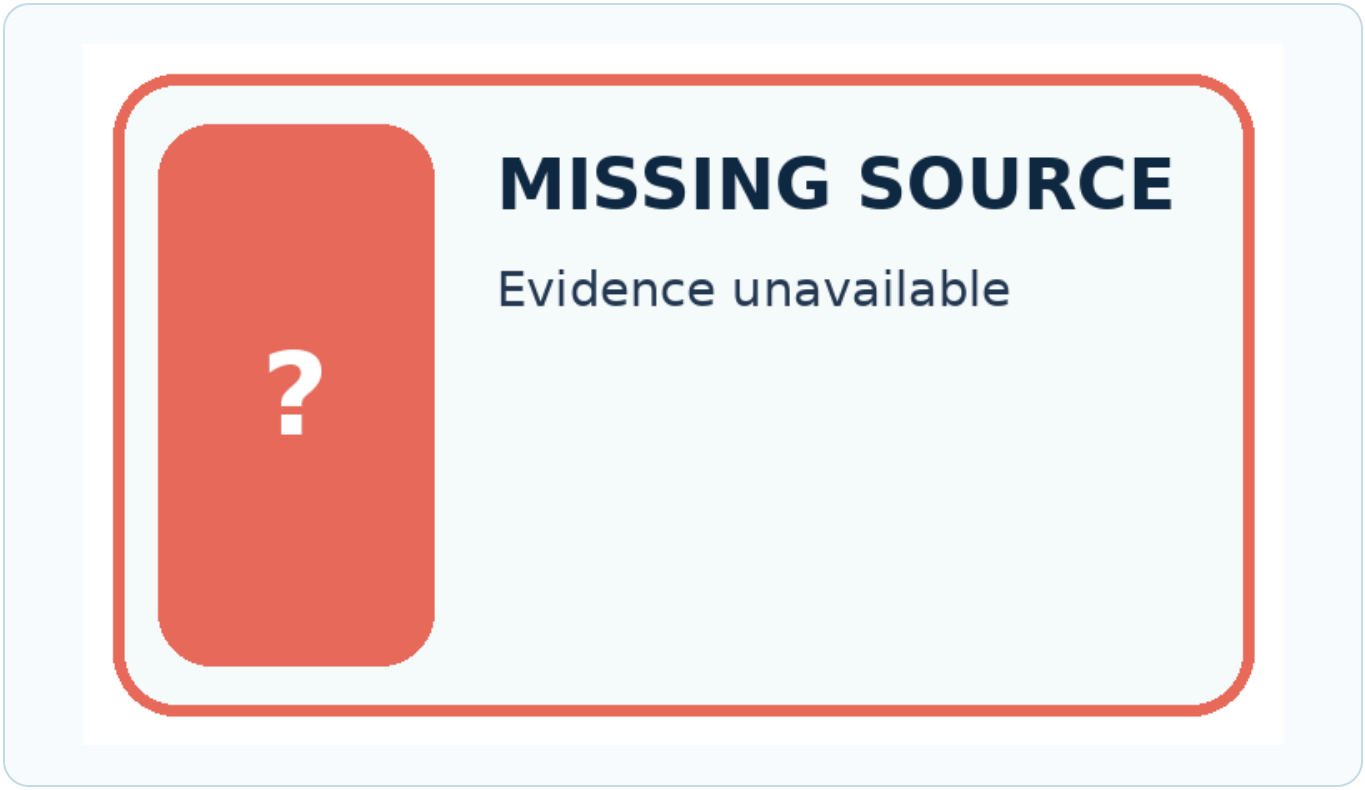
Four pictures with a purpose



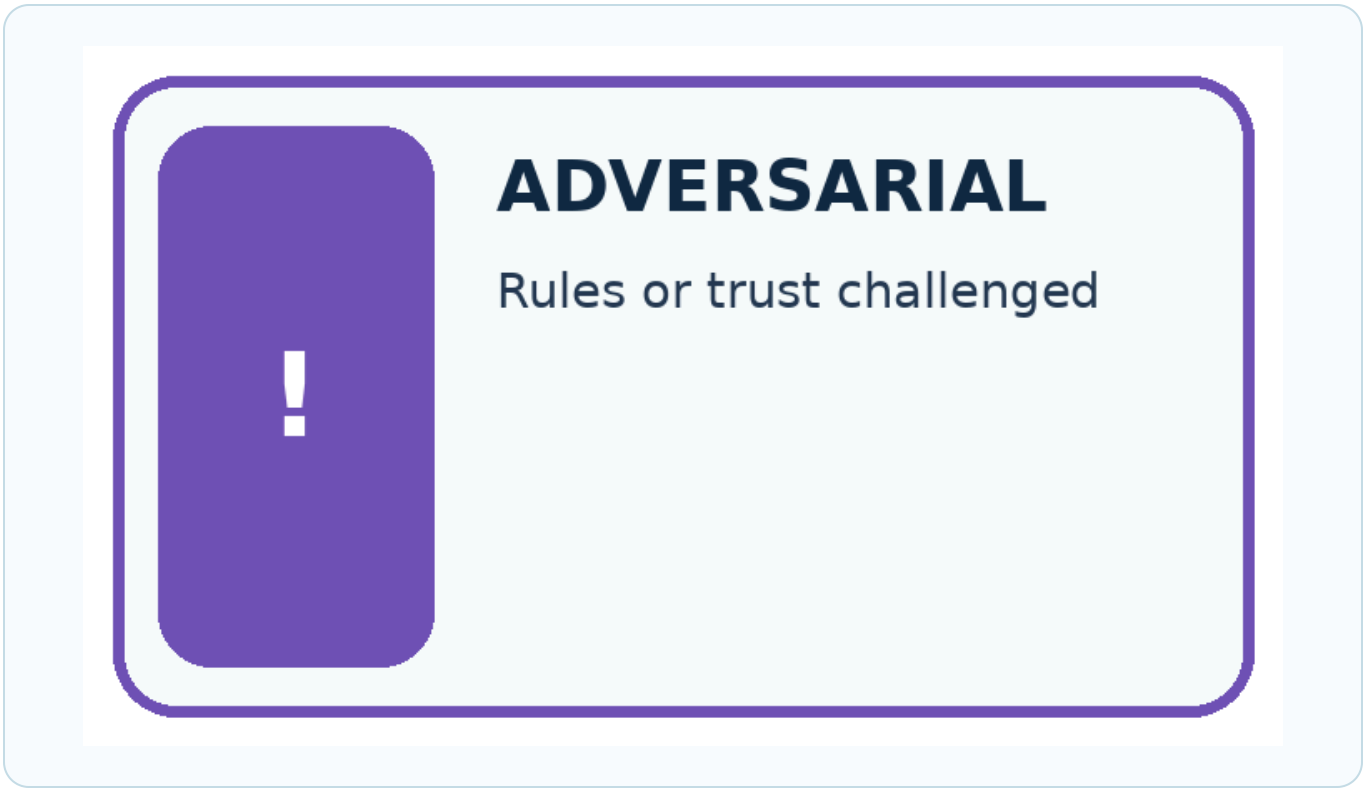
Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.



Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

> **Write or draw your reasoning and evidence.**

Predict the supplied changed case

- Change: LIB-v = 2 is unavailable.

▼ Check after predicting

Prediction: Report that the current time cannot be verified.

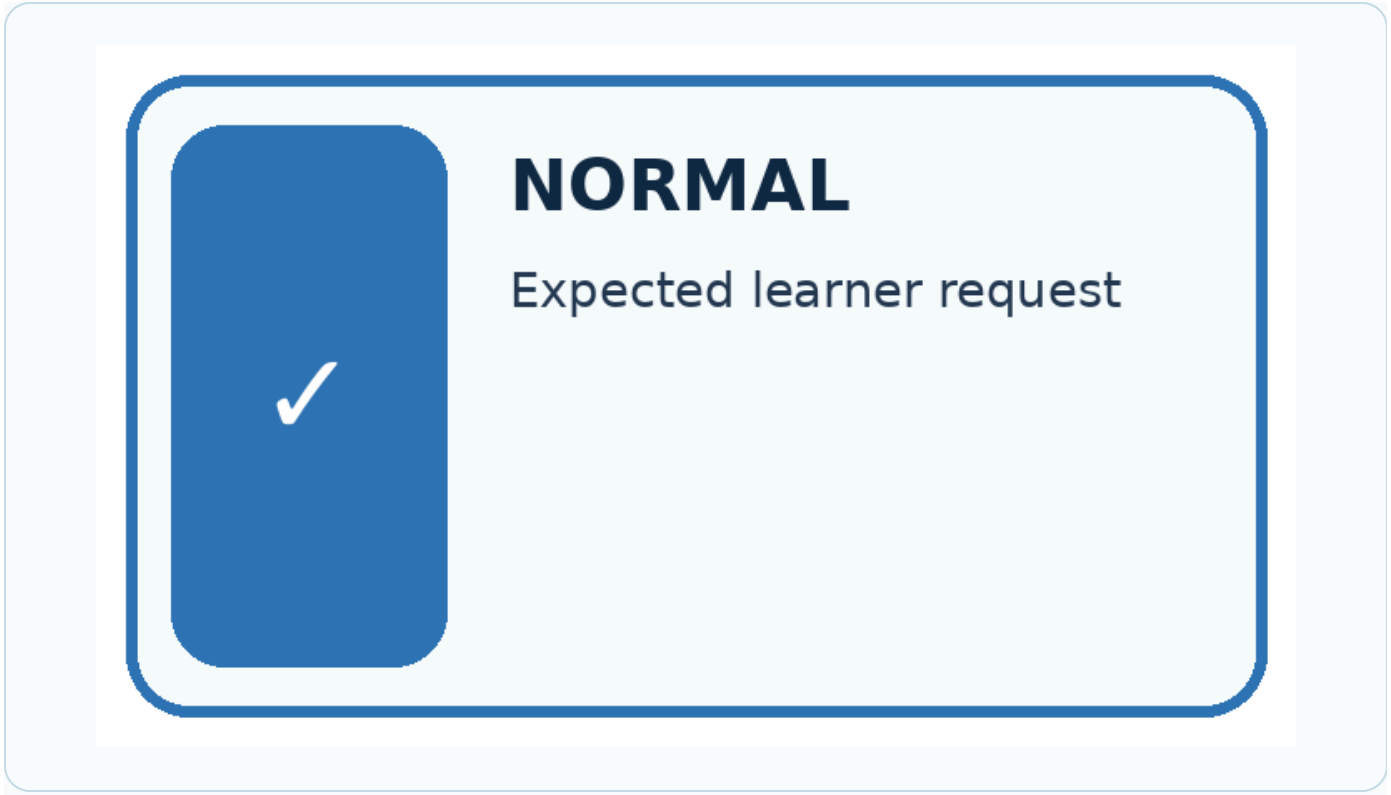
Reason: The unsupported old guess does not become current **evidence** when **retrieval** fails.

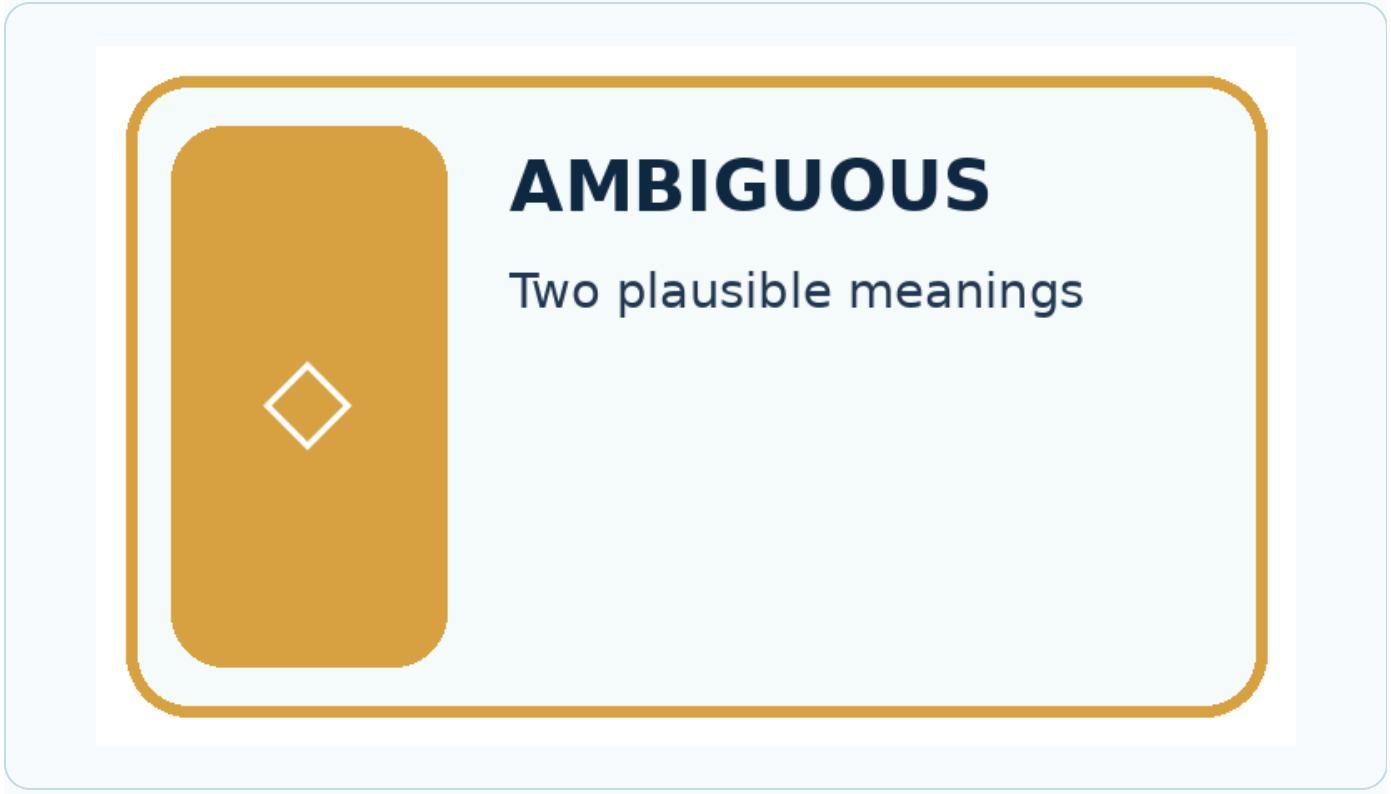
▼ Earlier wording, alternatives and picture transcripts

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.





NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Trace the library answer back to its current notice

> Keep the stale room-index review separately labelled.

Read the complete library record

The library assistant remembers an old opening time, but a current notice can be retrieved.

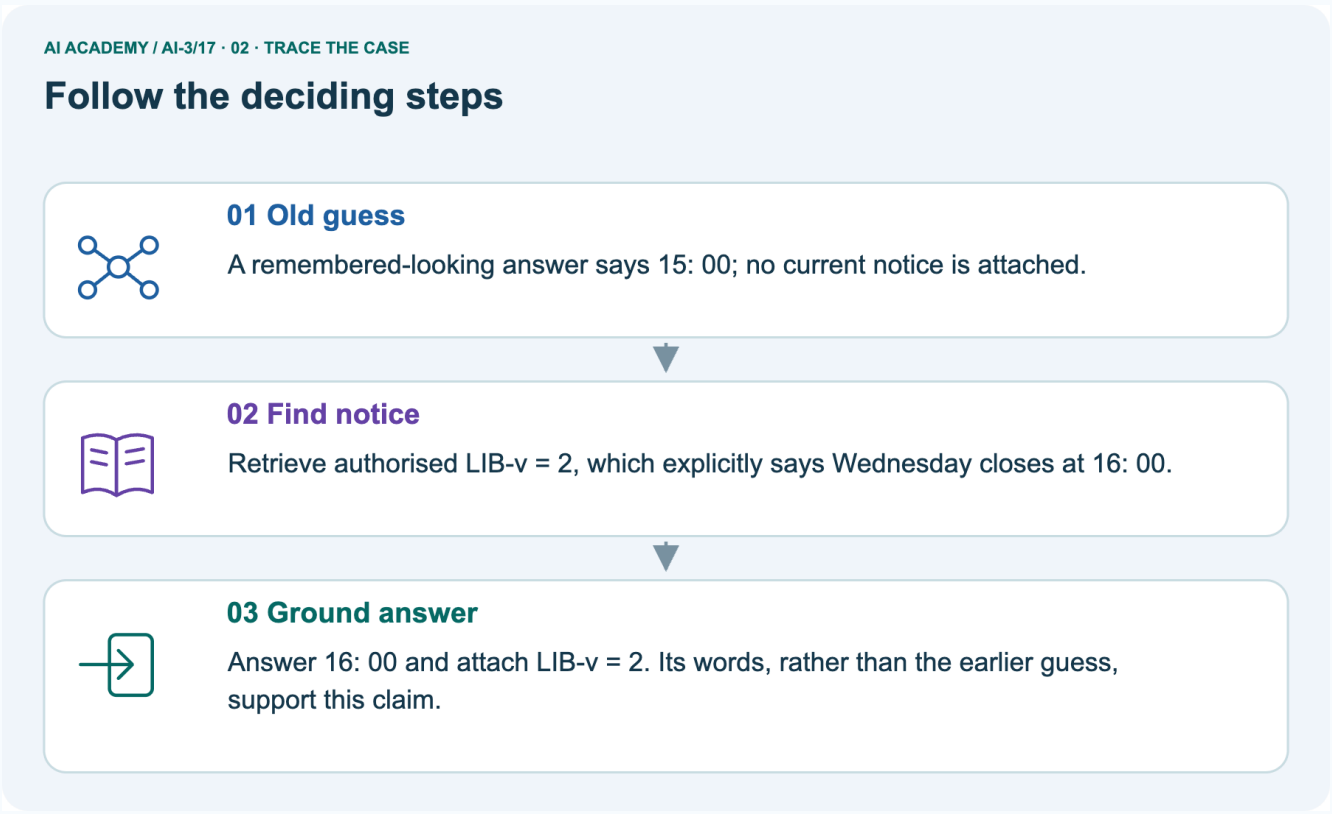
Approved notice LIB-v = 2 says Wednesday closes at 16: 00; the model previously produced 15: 00. The question asks the current Wednesday closing time.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Library case: LIB-v2 is explicitly supplied as the current approved notice for the requested Wednesday; it says closing time 16:00. The earlier model-produced 15:00 is a supplied fictional output, not evidence about actual model internals. In the changed case the notice is unavailable and no other current source is supplied.

Report that the current closing time cannot be verified; do not assert that 15:00 is therefore correct or false.

Trace notice to grounded answer



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Old guess	A remembered-looking answer says 15: 00; no current notice is attached.
2. Find notice	Retrieve authorised LIB-v = 2, which explicitly says Wednesday closes at 16: 00.
3. Ground answer	Answer 16: 00 and attach LIB-v = 2. Its words, rather than the earlier guess, support this claim.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

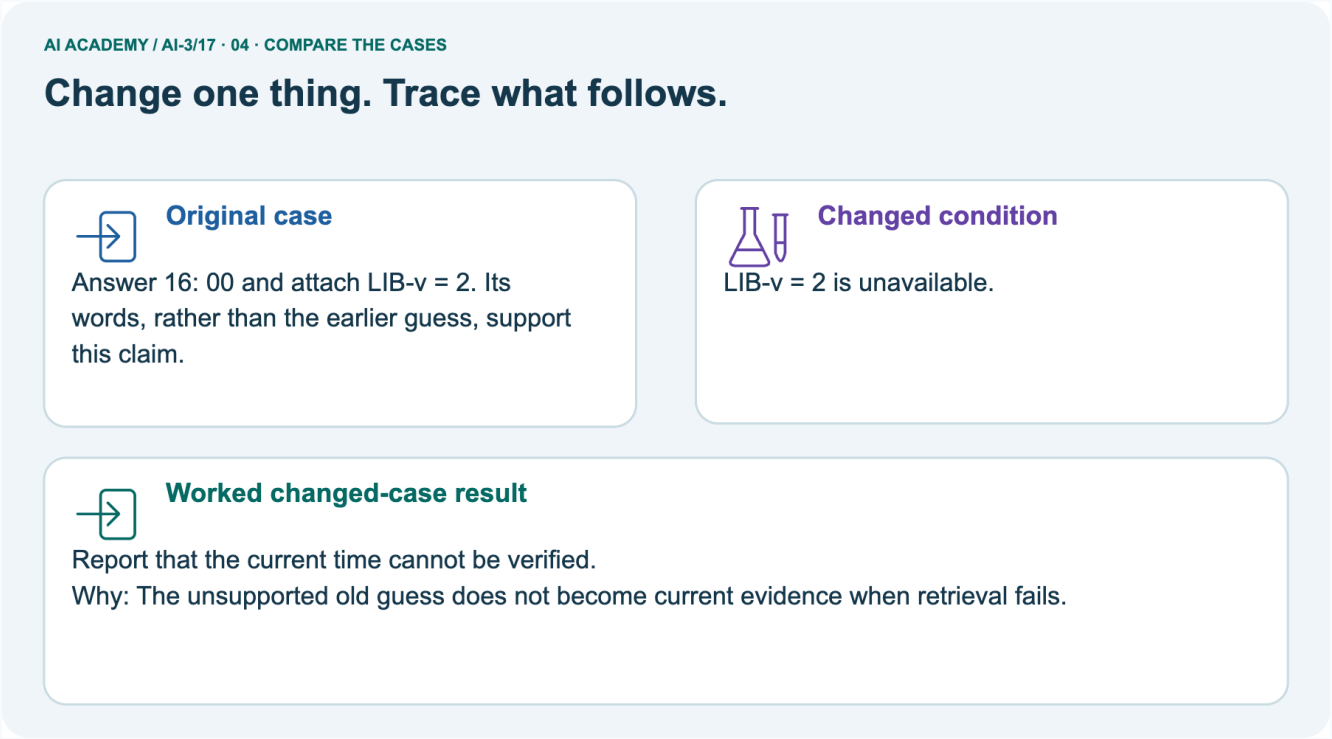
Read the picture in order. Answer 16: 00 and attach LIB-v = 2. Its words, rather than the earlier guess, support this claim. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Predict removal of the notice

- Change one thing: LIB-v = 2 is unavailable. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Name remaining source limits

- A retrieved document can itself be wrong or outdated; source approval, date and scope still need checking.

- Read the labelled picture: Why can the assistant not certify 15: 00 when the notice is missing? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

▼ Review · Separate stale room index

Index v1 contains notice N1 for room A. **Corpus** v2 supersedes it with N2 for room B, but the query still returns N1. What can be reported?

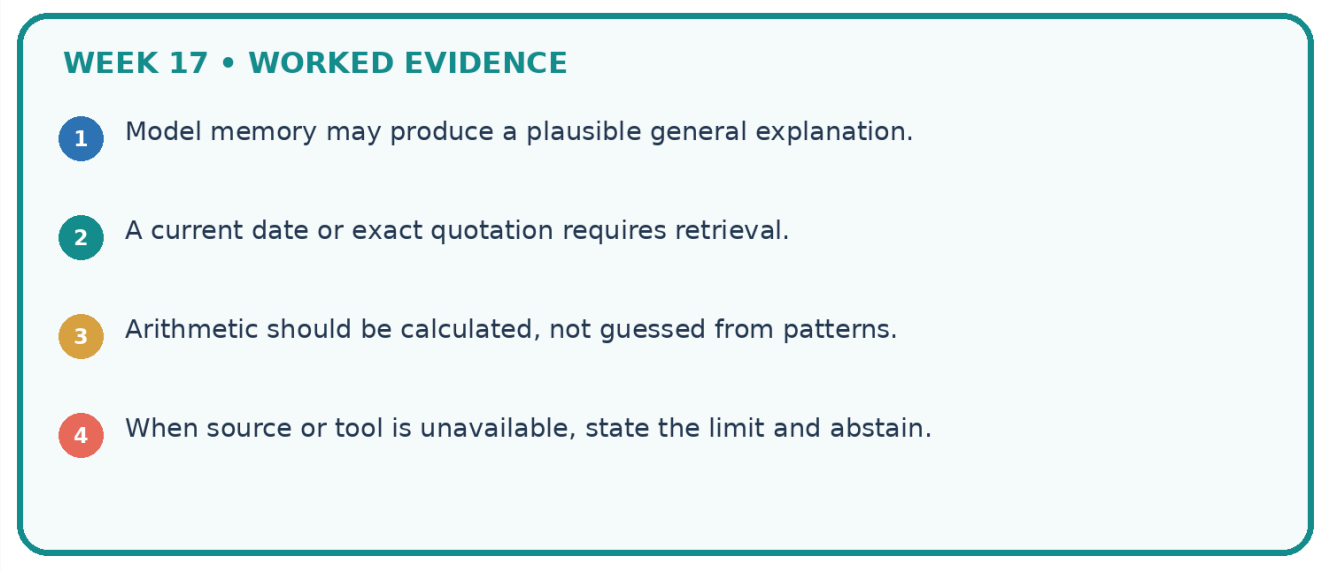
The stale **index** returns old **evidence**. It cannot establish current room approval; refresh against authorised **corpus** v2 or report unresolved **freshness**.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. Model memory may produce a plausible general explanation.

2. A current date or exact quotation requires **retrieval**.

3. Arithmetic should be calculated, not guessed from patterns.

4. When source or tool is unavailable, state the limit and **abstain**.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

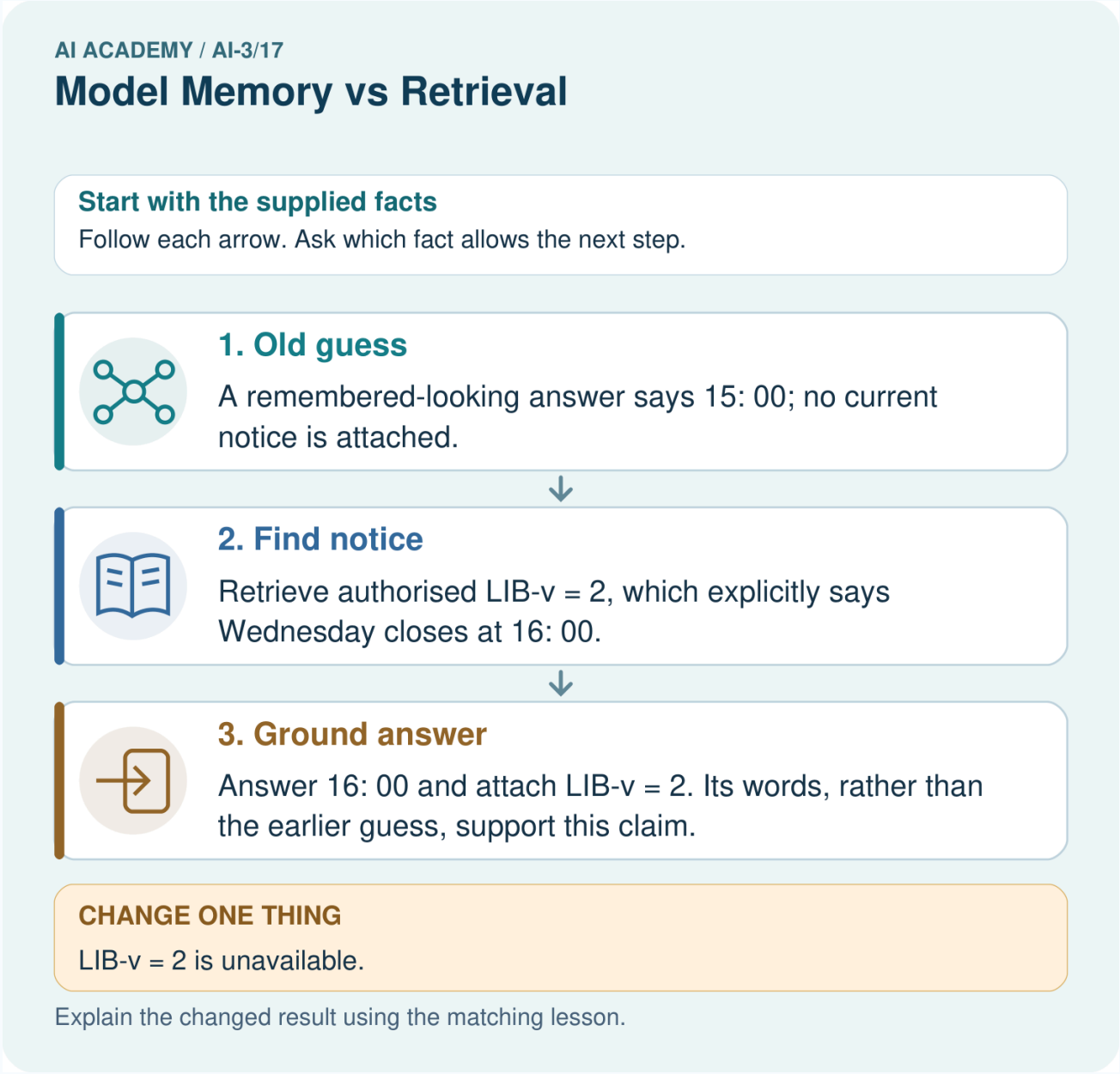
The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Model Memory vs Retrieval

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find the evidence shortcut

> Check which record actually supports the requested fact.

Repair confidence and search shortcuts

Claim to repair: The old answer sounds confident, so use it when retrieval fails. Missing current evidence does not upgrade generated wording into an inspected source. Conversely, a retrieved passage is not automatically current, authorised or relevant. In the stale room-index case, corpus v2 supersedes N1 with N2; retrieving N1 from index v1 cannot establish current approval.

Refresh against the authorised corpus or report the freshness gap.

Design a revealing follow-up

> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

▼ Earlier wording, alternatives and picture transcripts

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: A language model can inspect where every learned fact came from. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

> **Write or draw your reasoning and evidence.**

A closer explanation

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

MAIN LESSON · SECTION 8

Practise the stale-index trace with less help

> Keep room notices N1/N2 distinct from library notice LIB-v2.

The retrieval-similarity limit

High **retrieval** similarity does not override permissions or make an obsolete source current.

Complete the room-index W2

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “**Retrieval** and **index**ing”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Repair an all-future-cases claim

- Use the worked case for **Retrieval** and **index**ing. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Check your reasoning

Explain the missing library notice

- Why can the assistant not certify 15: 00 when the notice is missing? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Explain the picture

> **My explanation using a pictured step and the given facts**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Keep independent access work separate

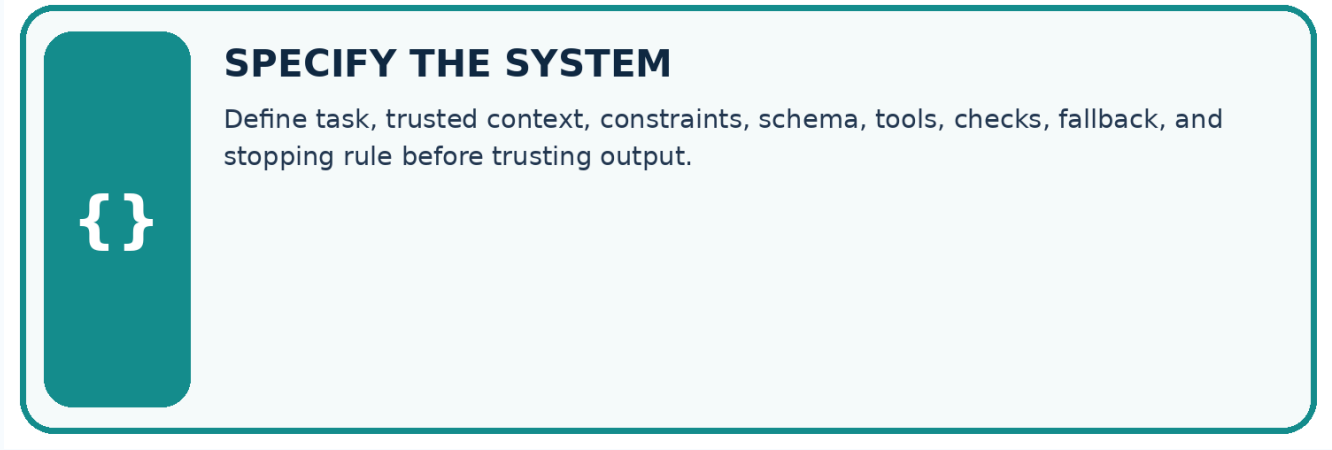
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, **retrieval**, tool calls, and code are inspectable parts of one AI product.



SET task, audience, constraints, **output**_schema LOAD approved_context or **retrieval**_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF **evidence** missing: ask, retrieve, or **abstain** RETURN useful **output** + source trail + limitations

Trace one **input**. Which line changes the state or chooses a path?

> **Write or draw your reasoning and evidence.**

A closer explanation

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

MAIN LESSON · SECTION 9

Route twelve complete requests by their evidence needs

Use supplied cards rather than inventing missing questions.

Predict before checking the route column

Question / supplied record	Expected route and evidence requirement
1 · Invent a clearly fictional library mascot.	Generate; label the invention, no factual source claim.
2 · Write a fictional welcome rhyme.	Generate; assess requested form without claiming real events.
3 · Current Wednesday library close; approved LIB-v2 says 16:00.	Retrieve/use LIB-v2; check scope and cite it.
4 · Current club hall; approved amendment assigns Hall 3 from day8; query day9.	Retrieve/use the amendment chain; answer Hall 3 under supplied policy.
5 · Pass rate from supplied 15/20 counts.	Calculate 75%; calculation does not verify measurement reliability.
6 · Supplied route legs 2 km + 500 m; 1 km = 1000 m.	Convert and calculate 2.5 km.

Question / supplied record	Expected route and evidence requirement
7 · Current library close; current notice unavailable, old output says 15:00.	Ask for the approved notice or state current time unverified.
8 · Intended meeting date; draft June31, no approved notice.	Flag invalidity and ask for the intended date.
9 · Student asks for restricted personal record R; no authorisation.	Do not retrieve or disclose R; offer an authorised route.
10 · Exact quotation from absent handbook H.	Request H; do not fabricate an exact quote.
11 · Room query retrieves N1/A, but supplied corpus policy says N2/B supersedes it.	Refresh eligible evidence; do not certify current approval from N1.
12 · Approved notice gives 18/24 passes; draft says 80%.	Retrieve/use the notice, then calculate 75%; routes can combine.

Record your route, source and fallback

1. Use these twelve complete fictional cards. Cover the expected-route column first. For each, name the requested output, evidence needed, eligible source, operation and fallback if evidence is missing. Then compare your reasoning with the supplied guidance. Generate, retrieve, calculate and ask/decline are not mutually exclusive: card12 combines retrieval with calculation.

   No real system or access control is tested here. An unavailable source is not proof that its claim is false. Keep the independent P/T/W case withheld.

Keep predictions and paper traces

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

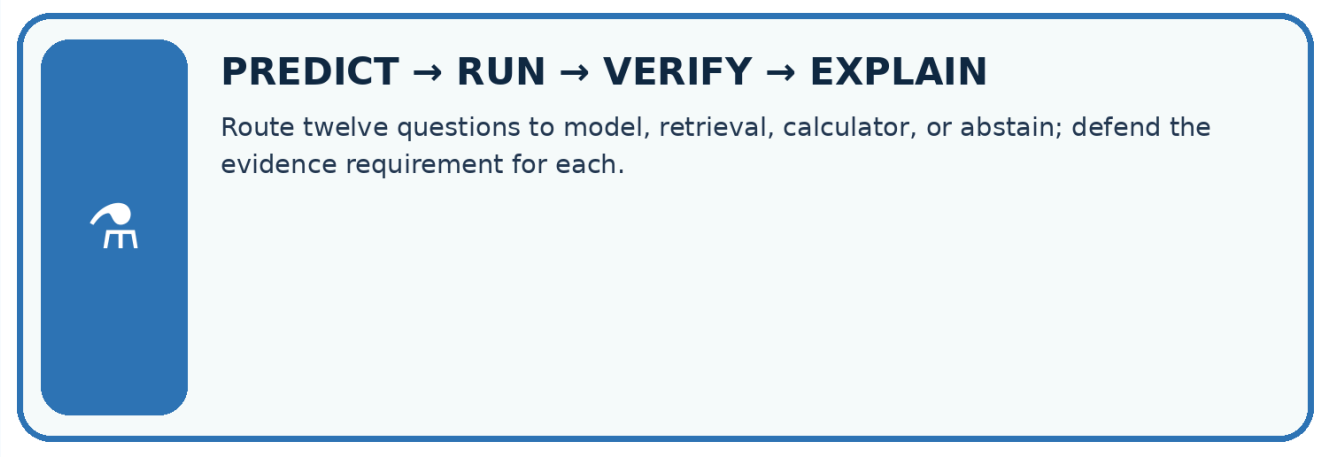
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Use the twelve supplied fictional request cards. Choose generation, **retrieval**, calculation or an **evidence**-seeking fallback; allow combined routes. Defend the exact source requirement and keep predicted routes separate from observed paper traces.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: inspect a fresh student retrieval result

> Use the stated record labels without inventing permissions.

Independent corpus and scope

Corpus has public P, teacher-only T and withdrawn W. A student search returns all three. Identify the filtering defect and whether hiding T after generation is sufficient.

Evaluate the supplied search result for a student request under the stated public, teacher-only and withdrawn labels. No real documents or live service are provided. Explain the defect and where the system must enforce the relevant conditions.

Your first attempt and help record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Apply a new case independently

Week 17 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A model recalls last year's opening hours; a retrieved current notice gives different hours. How should the answer use this **evidence**?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

The Study Buddy retrieves course passages and never presents model memory as a citation.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

A model states the library opens at 8 from learned patterns. A supplied current notice says 9. Explain the stronger answer and one remaining check.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Keep the project's evidence route inspectable

> A generated answer needs a source trail for source-dependent claims.

This week's contribution

For the Verified Study Buddy, record the query, requested time, corpus/index version, eligible source IDs, selected passage, calculation if any and final claim. Keep permissions before restricted evidence enters student-visible context. Do not present generated memory as an exact quotation or citation. If a current approved passage cannot be found, preserve the unresolved fact and request an authorised source.

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Retain the checkpoint and human choice

THIS WEEK'S CHECKPOINT The Study Buddy retrieves course passages and never presents model memory as a citation.

SYSTEM MAP Learner need → prompt/schema → approved sources/**retrieval** → model/tool → structured answer → verification → **human decision**

What will you add, change, or verify in the project this week?

> **Write or draw your reasoning and evidence.**

What should remain under **human control**?

> **Write or draw your reasoning and evidence.**

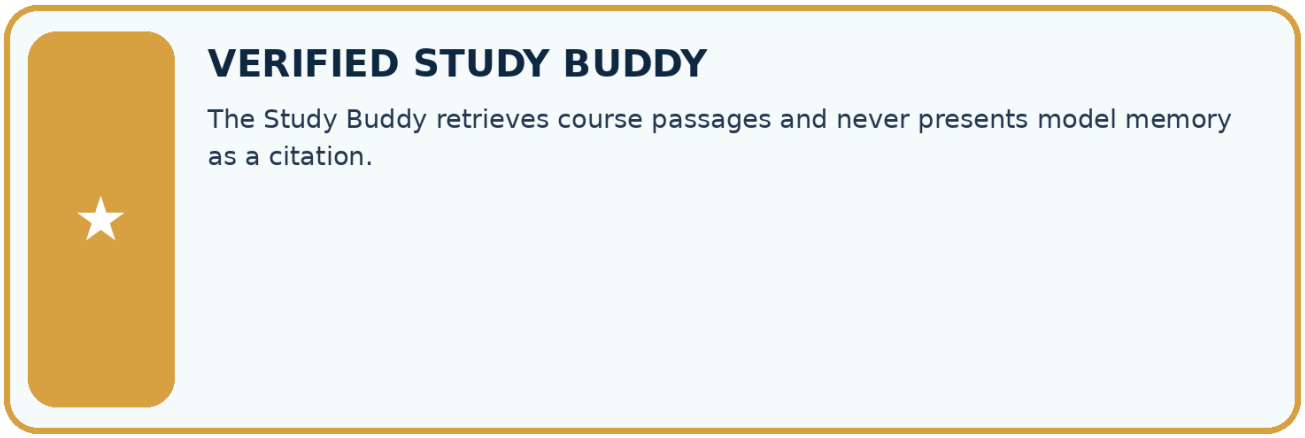
RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits.
Never confuse fluent **output** with verified truth.

▼ **Earlier wording, alternatives and picture transcripts**

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



Evidence for your project

MAIN LESSON · SECTION 16

Finish: name the source, operation and unresolved gap

> A fluent answer is not its own evidence.

Explain without notes

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of **retrieval** and **index**ing

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	My own example	Not this / boundary
parameters	Values within a model adjusted during training; distinguish them from settings chosen by a developer.	
retrieval	Finding and returning existing information from a source or collection.	
grounding	Connecting an answer or action to specified evidence or observations.	
abstain	Decline to make a prediction when the available evidence does not meet the decision rule.	

Terms for the connected explanation

Word	Meaning
Corpus	The collection of records available for a task
Index	A searchable representation of that collection

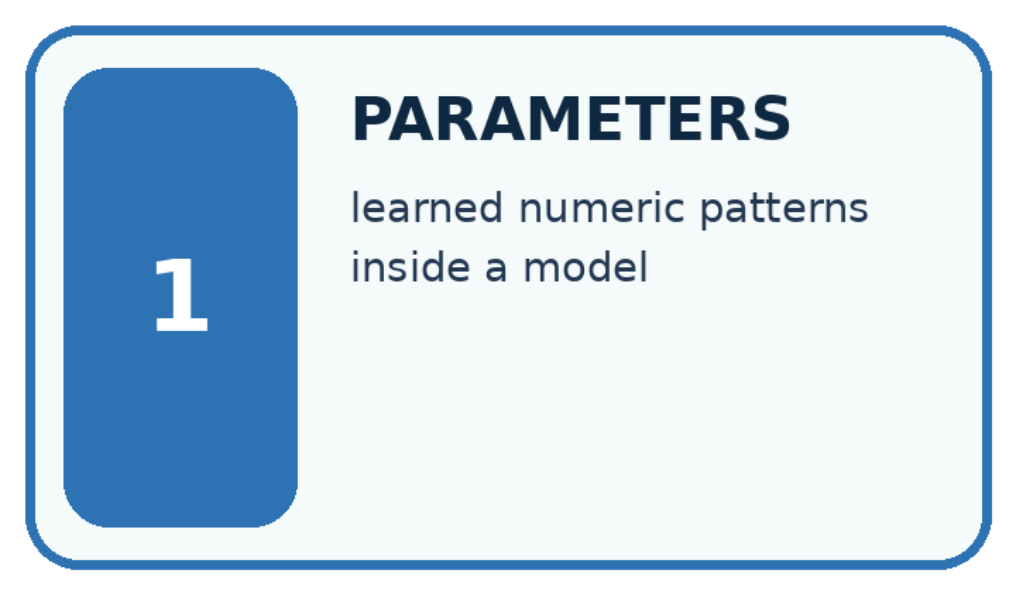
Word	Meaning
Freshness	Whether a record is current for the requested time

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

VOCABULARY LAB

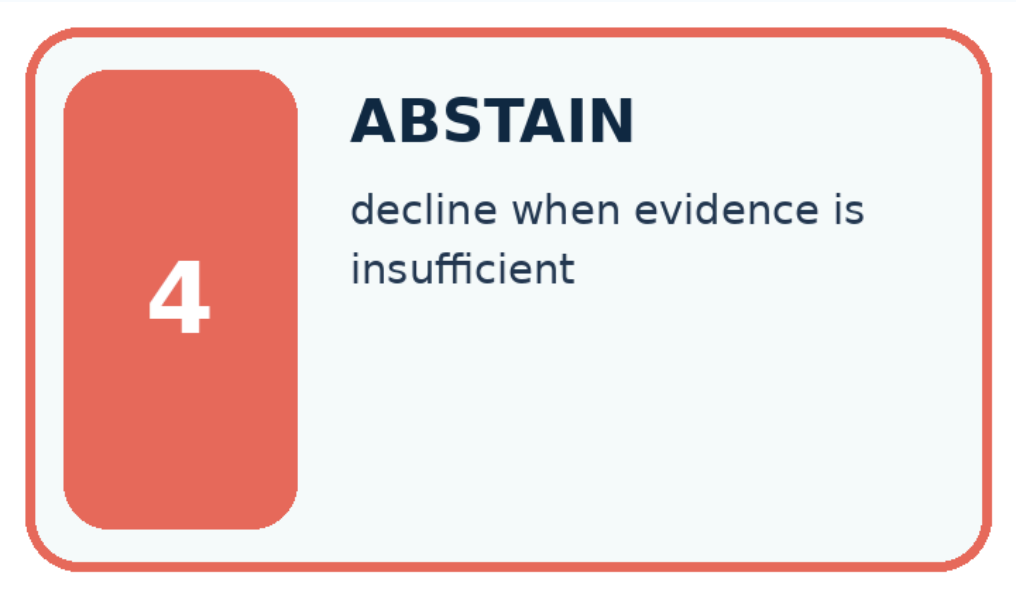
Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.



</details>


2
RETRIEVAL
search external stored sources
3
GROUNDING
tie an answer to supplied evidence



Words used in the closer explanation

▼ **Choose a stronger thinking challenge**

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

- My strongest **evidence** of independent understanding:

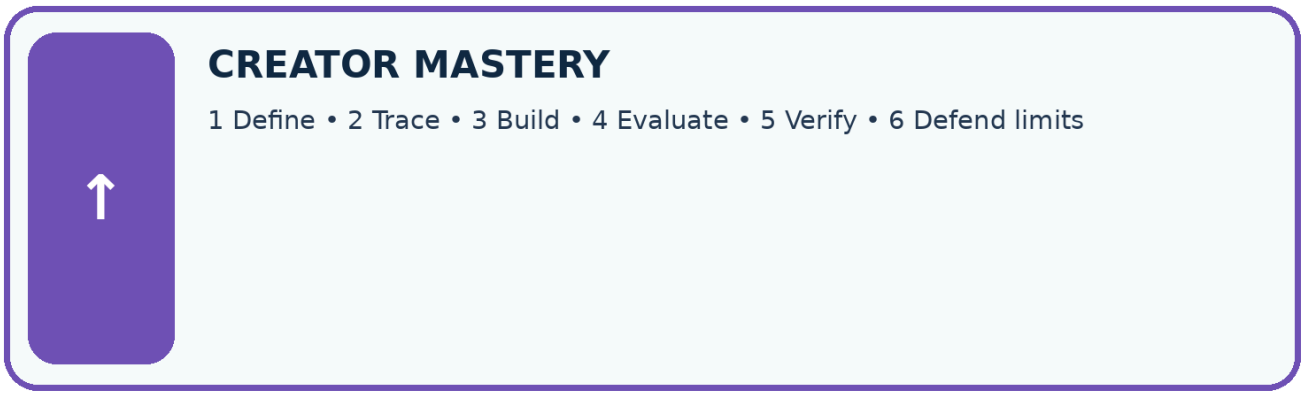
Write or draw your reasoning and evidence.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



▼ Reference: check your explanation

Reference: check your explanation

> Use after practice. Inspect a new student corpus result independently is the selected independent case.

Check the evidence and its limit

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a boundary or missing-data case):
- 3. One limitation, fairness concern, or situation needing human review:

> **Write or draw your reasoning and evidence.**

> **Write or draw your reasoning and evidence.**

Self-check and record

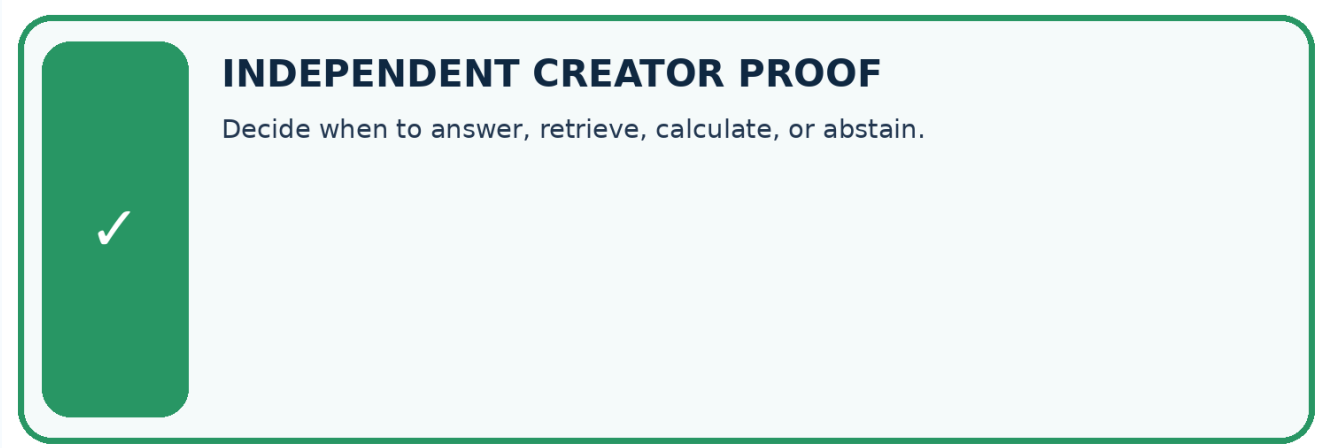
SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

▼ **Earlier wording, alternatives and picture transcripts**

INDEPENDENT MASTERY PROOF

Checklist for explaining this week's idea

Use after practice. Inspect a new student corpus result independently is the selected independent case.



EXPLANATION CHECK Require eligible active evidence and a defensible explanation of when filtering must occur.

Write or draw your reasoning and evidence.

▼ **Extra practice: Retrieval creates an inspectable evidence trail**

REFERENCE · SECTION 13

Extra practice: Retrieval creates an inspectable evidence trail

> Try the question before opening its worked explanation.

Read the supplied case and question

One assistant answers without search; another retrieves a handbook paragraph.

Compare **evidence** trails.

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

The first uses learned patterns without an inspected source. The second supplies a passage that can be checked. **Retrieval** still does not guarantee correct use.

Evidence or stage	Value or result
No retrieval	Answer from learned patterns
Retrieval	Inspect a handbook paragraph
Verification	Check the paragraph supports the answer

▼ Earlier wording, alternatives and picture transcripts

Week 17 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Retrieval creates an inspectable evidence trail

One assistant answers without search; another retrieves a handbook paragraph.

Compare **evidence** trails.

The first uses learned patterns without an inspected source. The second supplies a passage that can be checked. **Retrieval** still does not guarantee correct use.

▼ **Reference: original lesson checklist**

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Explain the mechanism and terminology of retrieval and indexing	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: The library assistant remembers an old opening time, but a current notice can be retrieved.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- The library assistant remembers an old opening time, but a current notice can be retrieved.
- Which labelled step matches this detail? Answer 16: 00 and attach LIB-v = 2. Its words, rather than the earlier guess, support this claim.
- 1. Old guess
- 2. Find notice
- 3. Ground answer

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

- After your attempt, compare with the supported explanation: Ground answer Answer 16: 00 and attach LIB-v = 2. Its words, rather than the earlier guess, support this claim. Read the picture in order.

  Answer 16: 00 and attach LIB-v = 2. Its words, rather than the earlier guess, support this claim. The arrows show which recorded **input** or intermediate result each decision uses.

  Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

Practice 2: explain the deciding evidence

Changed-case question

- Changed-case practice: LIB-v = 2 is unavailable. Which conclusion is supported by the supplied facts?

- 1. We can decide the result without examining the changed **input** or the procedure.

- 2. This one example guarantees correct results for every future **input**.

- 3. Report that the current time cannot be verified.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

- After your attempt, compare with the supported explanation: Report that the current time cannot be verified. The unsupported old guess does not become current **evidence** when **retrieval** fails.

> **My selected conclusion, deciding fact and explanation**
>
> ---
>
> ---
>
> ---
>
> ---

▼ **Check the changed case**

Repair a missing building block

Repair and save your work

Review Verify Numbers and Dates (ai-3/16). Goal: Verify a numeric claim using its source, denominator and units. Short practice: Which part can be corrected immediately from existing **evidence**, and which needs follow-up?. Return to this week and explain what you changed.

Review Hallucination Failure Types (ai-3/14). Goal: Explain the **mechanism** and terminology of generation failures and verification. Short practice: Can we conclude there will definitely be no lunch from S alone?.

Return to this week and explain what you changed.

Review Visual Composition (ai-3/11). Goal: Check object counts, spatial relations and arrow meaning separately. Short practice: Which checks pass after that single change, and what still needs repair?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Keep your completed work

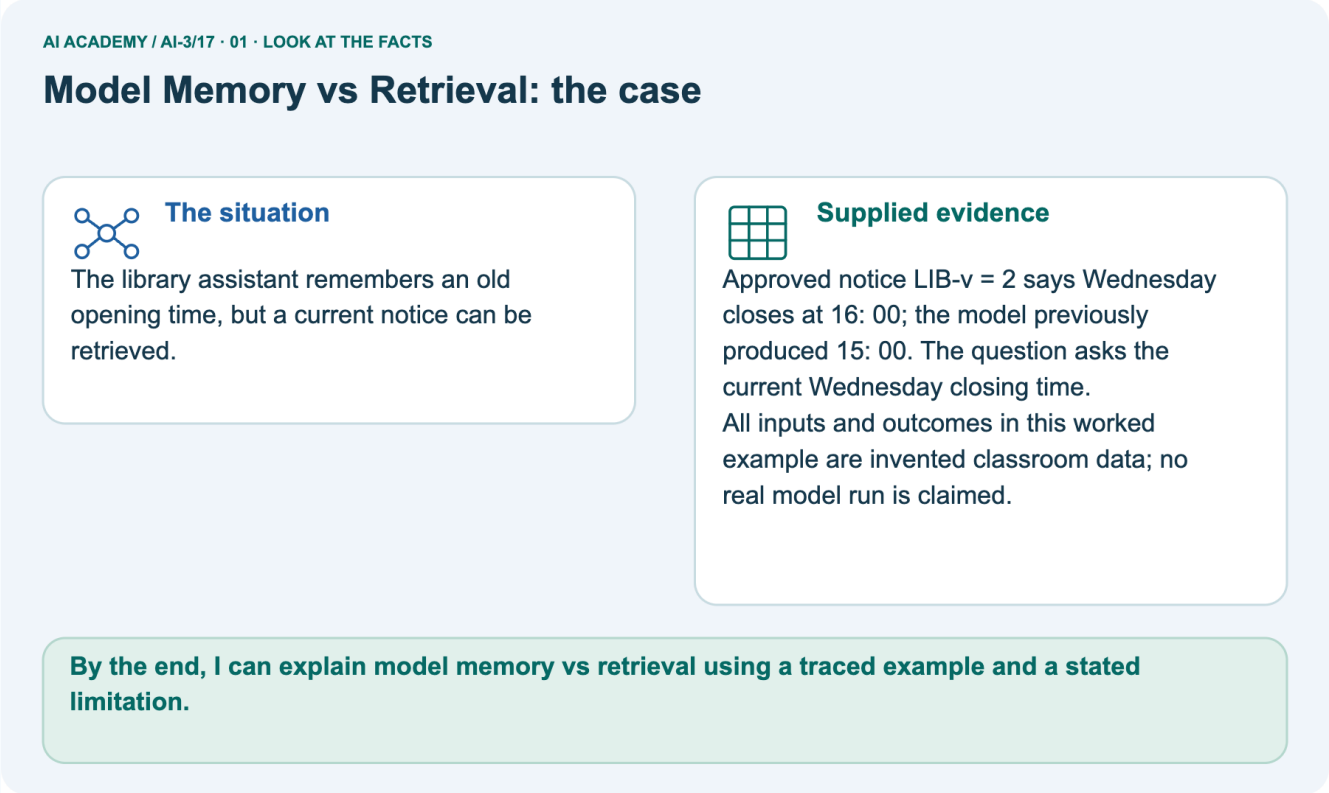
▼ **Reference: the six-picture journey**

REFERENCE · SECTION 18

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Model Memory vs Retrieval: the case



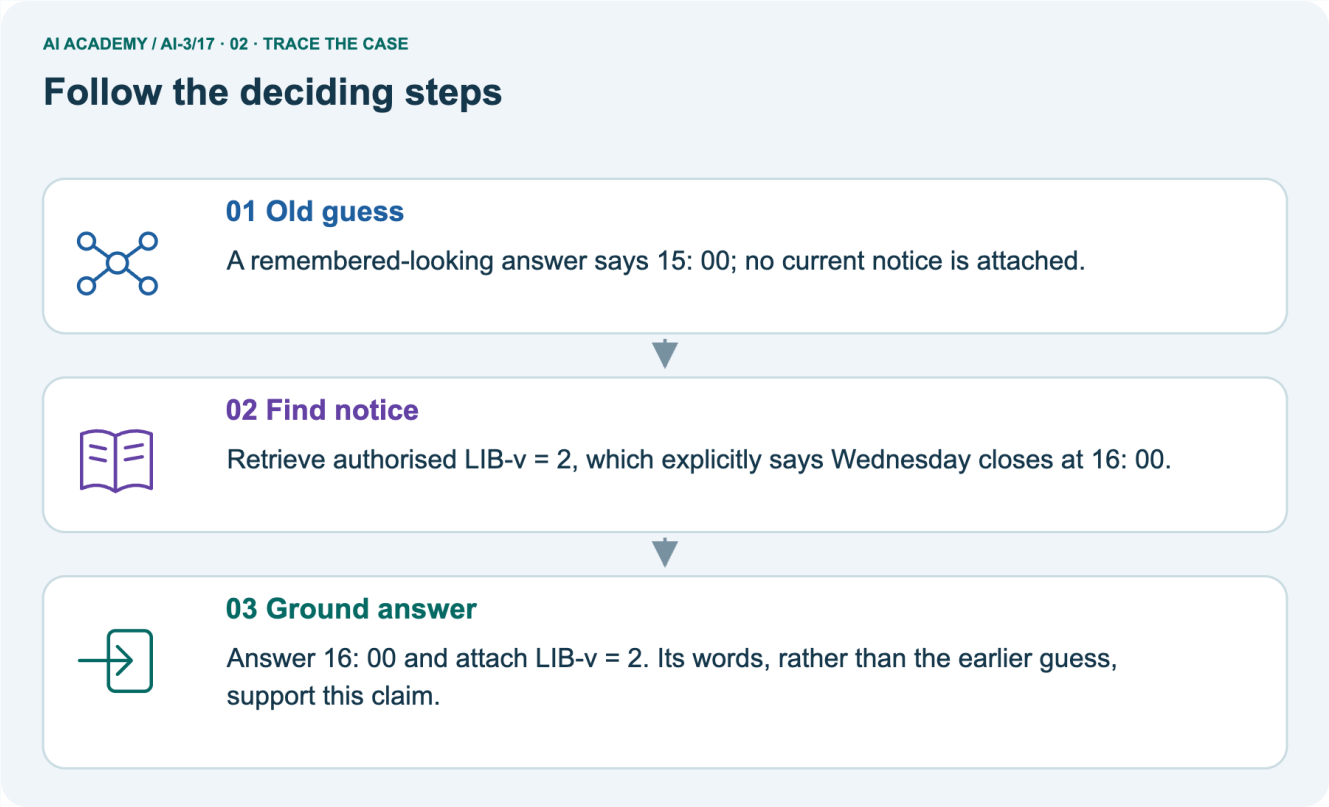
Follow the labelled picture with the matching explanation. Open the vector diagram

The library assistant remembers an old opening time, but a current notice can be retrieved. Supplied facts: Approved notice LIB-v = 2 says Wednesday closes at 16: 00; the model previously produced 15: 00.

The question asks the current Wednesday closing time. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



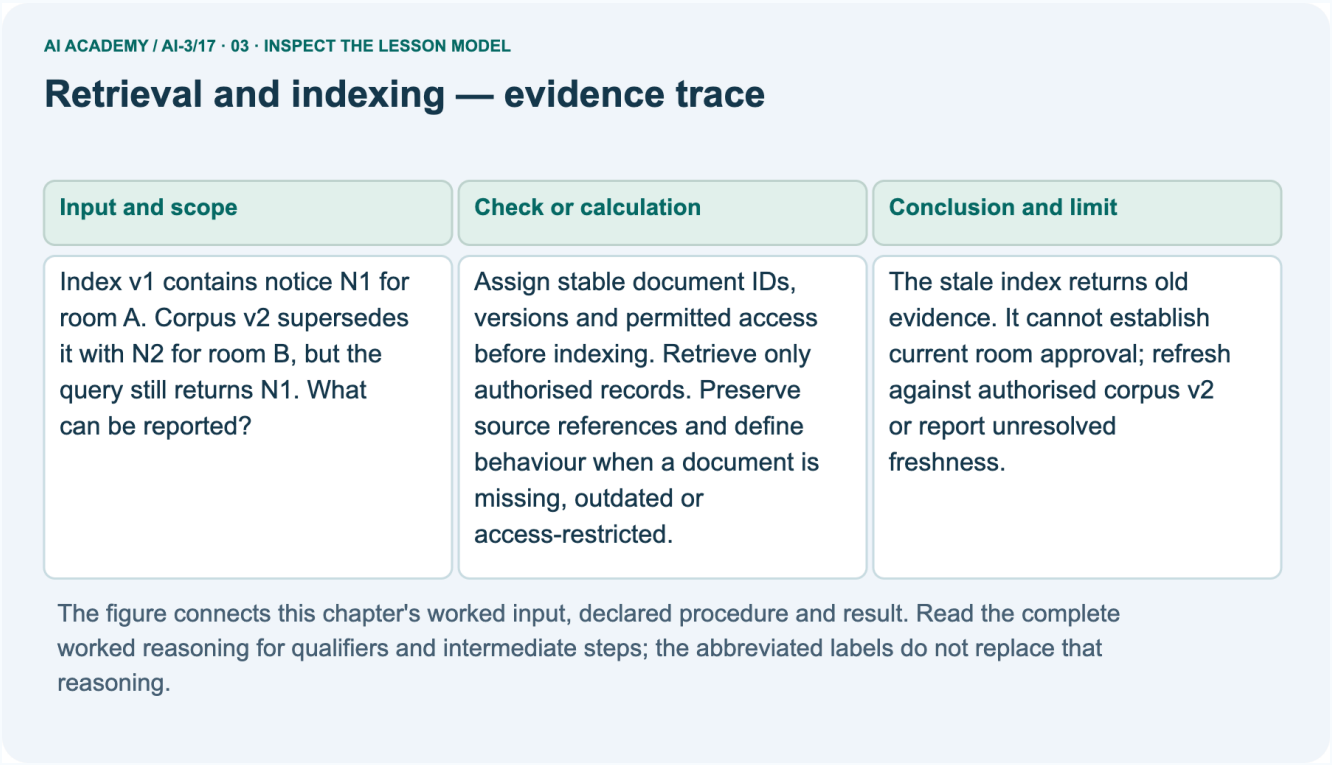
Follow the labelled picture with the matching explanation. Open the vector diagram

Old guess: A remembered-looking answer says 15: 00; no current notice is attached. → Find notice: Retrieve authorised LIB-v = 2, which explicitly says Wednesday closes at 16: 00. → Ground answer: Answer 16: 00 and attach LIB-v = 2.

Its words, rather than the earlier guess, support this claim.

Point to each numbered stage and explain which recorded **input** it uses.

03. Retrieval and indexing — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Retrieval and **index**ing — **evidence** trace. **Index** v1 contains notice N1 for room A. **Corpus** v2 supersedes it with N2 for room B, but the query still returns N1.

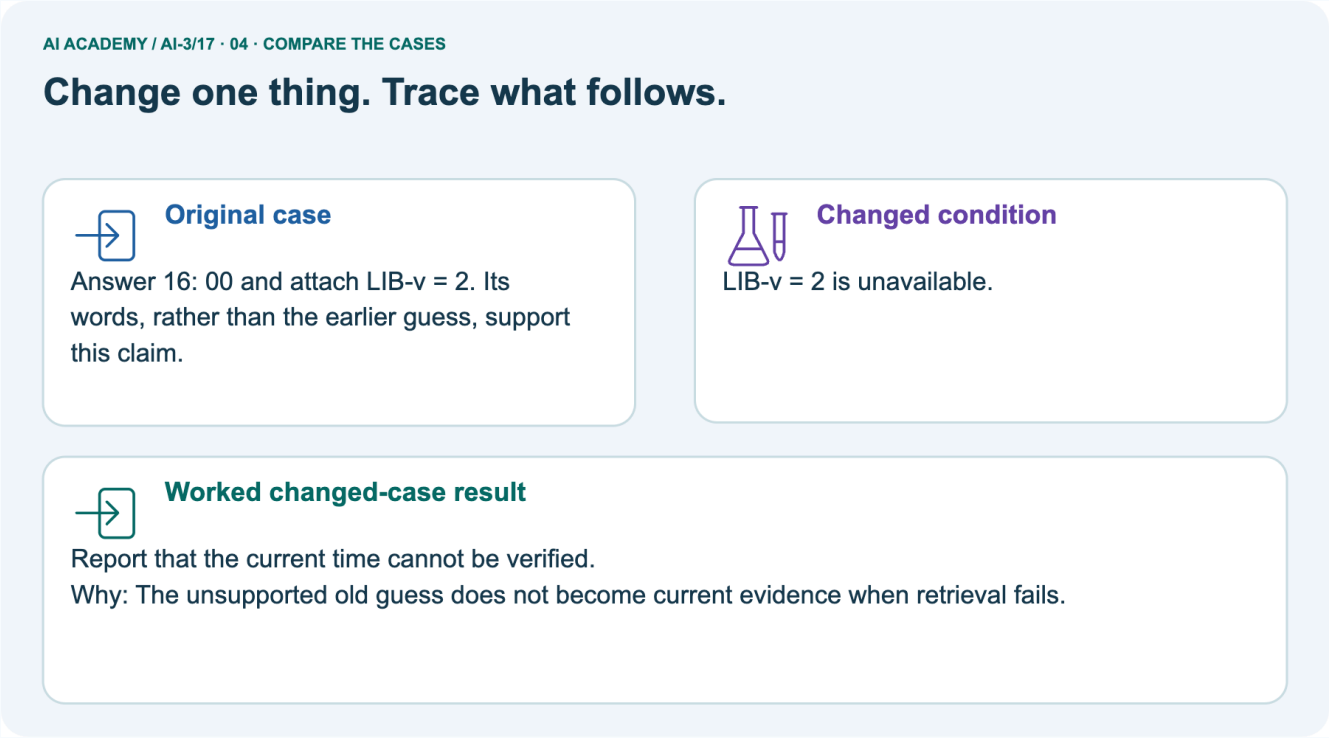
What can be reported? | Assign stable document IDs, versions and permitted access before **index**ing. Retrieve only authorised records.

Preserve source references and define behaviour when a document is missing, outdated or access-restricted. | The stale **index** returns old **evidence**.

It cannot establish current room approval; refresh against authorised **corpus** v2 or report unresolved **freshness**.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



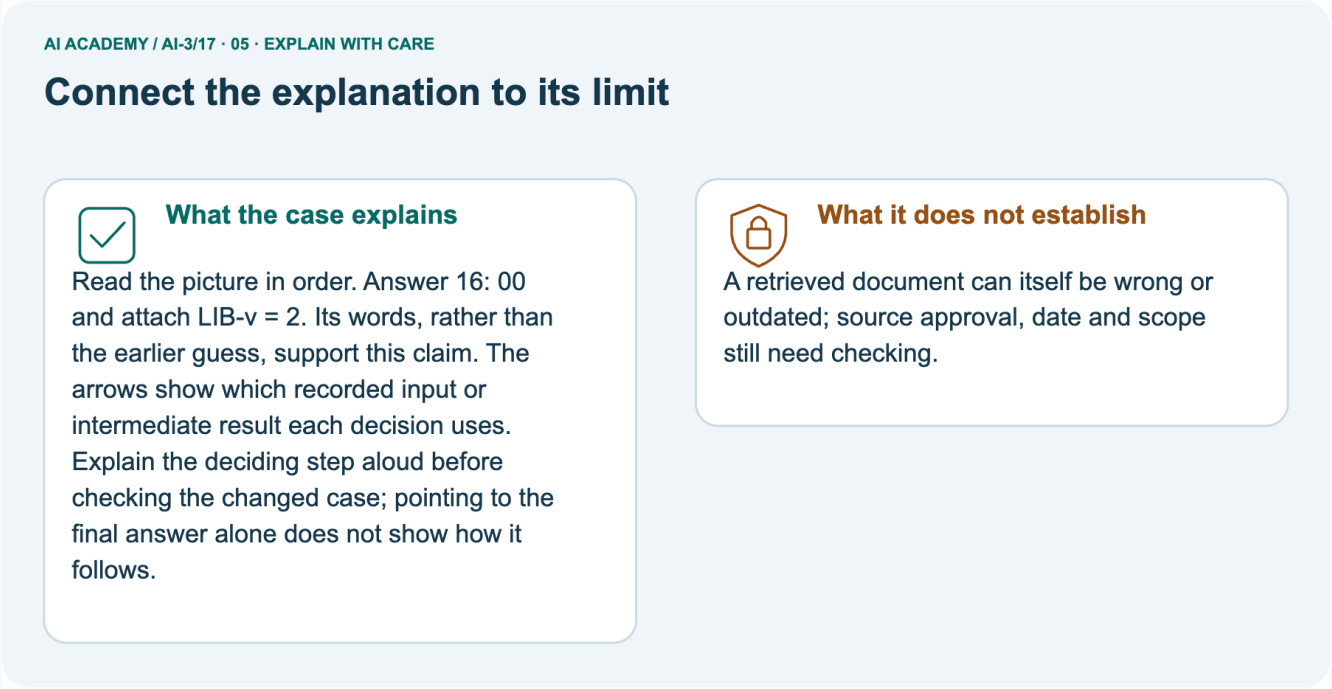
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Answer 16: 00 and attach LIB-v = 2. Its words, rather than the earlier guess, support this claim. Changed condition: LIB-v = 2 is unavailable. Expected result: Report that the current time cannot be verified.

Reason: The unsupported old guess does not become current **evidence** when **retrieval** fails.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

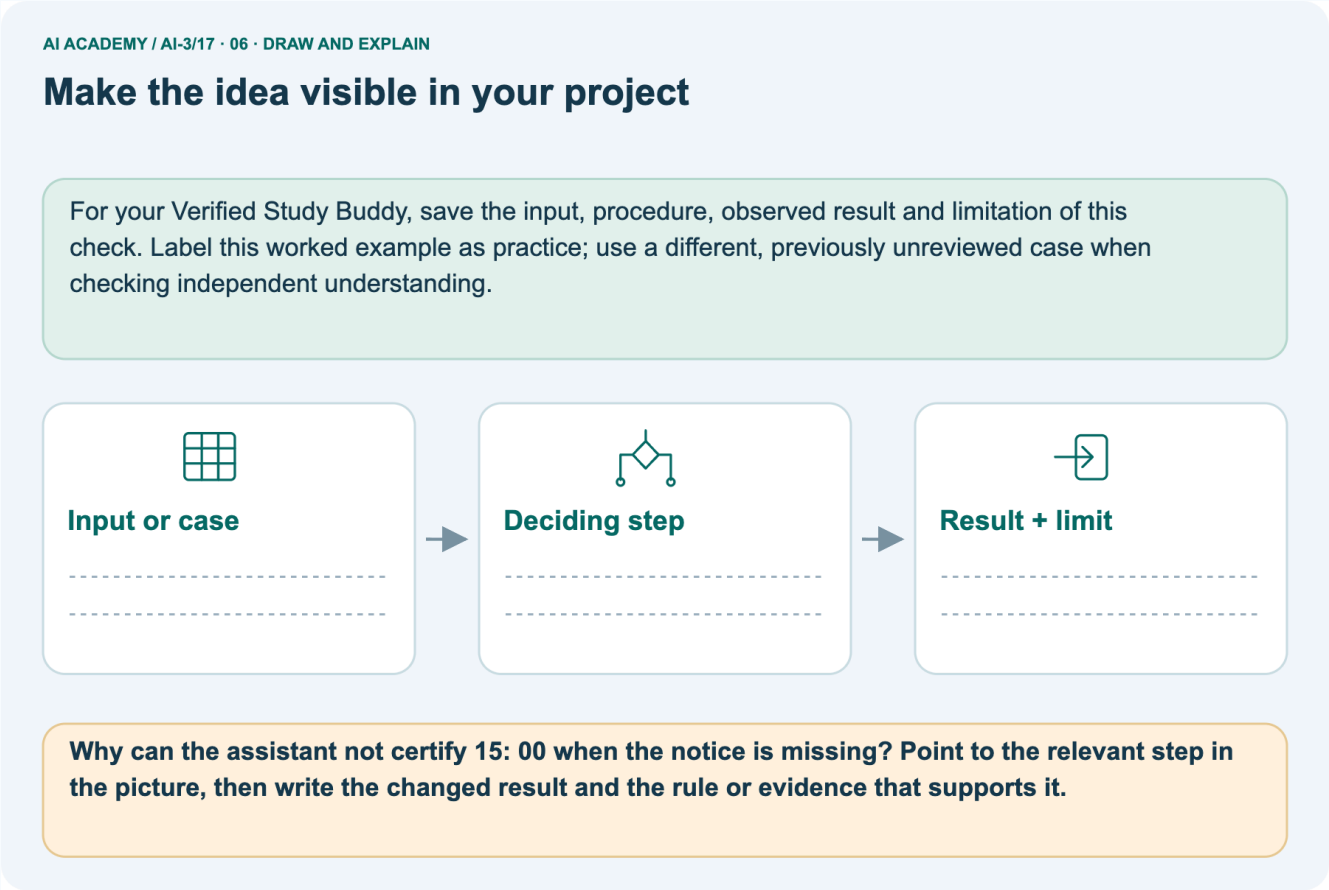
Read the picture in order. Answer 16: 00 and attach LIB-v = 2. Its words, rather than the earlier guess, support this claim. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A retrieved document can itself be wrong or outdated; source approval, date and scope still need checking.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why can the assistant not certify 15: 00 when the notice is missing? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Model Memory vs Retrieval: the case

02. Follow the deciding steps

03. Retrieval and indexing — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.